

Solution Architecture

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TeamID	XXXXXX
Project Name	EduGenie-Google Gemini Powered Learning Assistant
Maximum Marks	5Marks

Solution Architecture Diagram

The EduGenie-Google Gemini Powered Learning Assistant follows a layered architecture Consisting of the Presentation Layer, Application Layer, AI Processing Layer, and Data Storage Layer. Students interact with the web interface to submit learning requests. The Fast API server processes these requests and communicates with Google Gemini 1.5 Pro and LaMini-Flan-T5 models to generate educational responses. The generated results are stored and displayed through the application interface.

Presentation/Client Layer

HTML5 • CSS3 • JavaScript • Responsive
Web UI



APIGateway/Server

Python • FastAPI • Uvicorn



Authentication

User Login •
Session Management

Core Logic

Concept
Explanation •
Question Answering
• Quiz Generation
Text Summarization
• Personalized

External APIs

Google Gemini 1.5
Pro • LaMini-Flan-T5



Data/Storage Layer

JSONFiles • UserHistory • LearningRecords • LocalStorage

Component Description Table

Component Name	Description/Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for students and teachers, such as concept explanation, Q&A, quiz Generation and learning recommendations.	HTML, CSS, Java Script
APIGateway	Receives requests from the frontend, validates them, and returns generated responses.	Python
CoreLogic Service	Processes user requests and manages content generation,summarization,andpersonalized recommendations.	Python, Fast API
Database/ Storage	Storesuserinformation,learninghistory,generated recommendations, and application data.	JSON